

Alathon L4904

High Density Polyethylene

Product Description

Alathon L4904 is a bimodal, high molecular weight, high density polyethylene resin with excellent processing characteristics. L4904 is selected by customers for pressure pipe applications including gas distribution, industrial piping, mining, oil & gas gathering, municipal water service lines and sewers. Customers typically use L4904 in applications requiring high resistance to pipe failure by rapid crack propagation and slow crack growth mechanisms. When L4904 is combined with a LyondellBasell approved color or black masterbatch at the correct use level, this compound may meet the following standards or requirements (see Technical Data Sheet for limitations and more information):

- ASTM D2513-09a for Polyethylene (PE) Gas Pressure Pipe, Tubing and Fittings
- ASTM D3350 Cell Classifications: **PE445574C CC3, PE445574E CC0, PE445576C CC3, and PE445576E CC0**
- CSA B137.1 for Pipe, Tubing, and Fittings for Cold-Water Pressure Services
- CSA B137.4 for Polyethylene Piping Systems for Gas Services: PE 4710 PLUS & PE 100
- NSF Standard 14 and Standard 61 for Potable Water Pipe and Fittings
- NSF Standard 14 for Gas Distribution Applications
- NSF Standard 358-1 for PE Pipe and Fittings for "Geothermal" Heat Pump Systems
- Plastics Pipe Institute (PPI) PE 4710 and PE100 per PPI TR-3

Regulatory Status

For regulatory compliance information, see Alathon L4904 [Product Stewardship Bulletin \(PSB\) and Safety Data Sheet \(SDS\)](#).

Status	Commercial
Availability	Asia-Pacific; North America; South & Central America
Application	Drinking Water Pipe; Gas Pipe
Market	Industrial, Building & Construction; Pipe
Processing Method	Pipe

Typical Properties	Nominal Value	English Units	Nominal Value	SI Units	Test Method
Physical					
Melt Flow Rate					
(190 °C/2.16 kg)	0.04	g/10 min	0.04	g/10 min	ISO 1133-1
(190 °C/2.16 kg)	0.04	g/10 min	0.04	g/10 min	ASTM D1238
(190 °C/5.0 kg)	0.18	g/10 min	0.18	g/10 min	ISO 1133-1
(190 °C/5.0 kg)	0.18	g/10 min	0.18	g/10 min	ASTM D1238
(190 °C/21.6 kg)	7.0	g/10 min	7.0	g/10 min	ASTM D1238
(190 °C/21.6 kg)	7.0	g/10 min	7.0	g/10 min	ISO 1133-1
Density					
(23 °C)	0.949	g/cm ³	0.949	g/cm ³	ASTM D1505
(23 °C)	0.949	g/cm ³	0.949	g/cm ³	ISO 1183-1
Mechanical					
Flexural Modulus					
(1% Secant)			1030	MPa	ISO 178
(2% Secant)	146000	psi			ASTM D790
Tensile Modulus, (23 °C)			750	MPa	ISO 527-1, -2

Tensile Stress at Break (23 °C)	5100 psi	30 MPa	ASTM D638 ISO 527-1, -2
Tensile Stress at Yield (23 °C)	3500 psi	24.5 MPa	ASTM D638 ISO 527-1, -2
Tensile Elongation at Break (23 °C)	615 %	615 %	ASTM D638 ISO 527-1, -2
PENT on Natural Resin, (2.4 MPa, 80 °C)	10000 hr	10000 hr	ASTM F1473
Values were determined on natural L4904 resin.			
PENT on Black Compound, (2.4 MPa, 80 °C)	5000 hr	5000 hr	ASTM F1473
Values were obtained from L4904 compounded with an approved masterbatch.			
Thermal			
Low Temperature Brittleness, F ₅₀	<-105 °F	<-76 °C	ASTM D746
DSC Induction Temperature	500 °F	260 °C	ASTM D3350
Oxidative-Induction Time, (200 °C)	100 min	100 min	ASTM D3895
Values were determined on L4904 compounded with an approved masterbatch.			
Conformance Testing			
Hydrostatic Design Basis			
(73 °F)	1600 psi	1600 psi	ASTM D2837
(140 °F)	1000 psi	1000 psi	ASTM D2837
Minimum Required Strength, (20 °C)	10 MPa	10 MPa	ISO 12162
Creep Rupture Strength, (20 °C, 12.4 MPa)	>200 hr	>200 hr	ASTM D1598
Resistance to Rapid Crack Propagation, Pc @ 32 °F	>12 bar	>12 bar	ISO 13477
Values were obtained on 8" SDR11 pipe made with L4904 and an approved masterbatch.			
Resistance to Rapid Crack Propagation, Tc @ 5 bar	<10 °F	<-12 °C	ISO 13477
Values were obtained on 4" SDR11 pipe made with L4904 and an approved masterbatch.			
Notched Pipe Test, (80 °C, 4.6 MPa)	>2500 hr	>2500 hr	ISO 13479
Values were obtained on 4" SDR11 pipe made with L4904 and an approved masterbatch.			

Notes

Typical Property values were determined on natural L4904 resin, unless otherwise noted. Conformance Test values were determined from L4904 compounded with an approved masterbatch.

These are typical property values not to be construed as specification limits.

Processing Techniques

Specific recommendations for resin type and processing conditions can only be made when the end use, required properties and fabrication equipment are known.

Approved Black Masterbatches

The following black masterbatches are approved for use with *Alathon* L4904 when added to produce a carbon black level in the final article of 2.00 - 2.56 wt%. When using a nominal 35 wt% carbon black masterbatch, this will result in a use level of the masterbatch between 6.0 and 7.3 wt%. Addition levels may need to be adjusted due to variation in the certified carbon black level in the particular masterbatch lot.

<u>Manufacturer</u>	<u>Masterbatch Code</u>
Ampacet Corporation	190872, 190872A
Chevron Phillips Chemicals Company LP	M368
Ingenia Polymers Corporation	IPBK019A, IPBK033A
Modern Dispersions Incorporated	PE 535-42
PolyOne Corporation	B60054, B60054C, PE-335-4710L BLACK PEC
RTP Company	PE 42000 BK MB, PE 42008 BK MB

Approved Color Masterbatches

<u>Manufacturer</u>	<u>Masterbatch Code</u>	<u>Use Level</u>	<u>Restrictions</u>
PolyOne Corporation	E002974 Grey	2%	non-potable water only

Chlorine Classification

Alathon L4904 meets the ASTM D3350-14 chlorine classification value of CC3 when combined into pipe or fittings at a carbon black level of 2.00 - 2.56 wt% using one of the approved black masterbatches listed below.

<u>Manufacturer</u>	<u>Masterbatch Code</u>
Ampacet Corporation	190872
Ingenia Polymers Corporation	IPBK033A
Modern Dispersion Incorporated	PE 535-42
PolyOne Corporation	PE-335-4710L BLACK PEC
RTP Company	PE 42000 BK MB

Caution

If an approved masterbatch is not used in the manufacture of pipe or fittings, the customer assumes all responsibility for determining that the final article meets the applicable material and pipe or fittings standards.

Chemical Resistance

Chemical resistance testing was conducted in accordance with ASTM D2513-01a, Section 5.4 on L4904 with an approved black masterbatch. Results can be found in the table below.

Reagent	Weight Change		Change in Tensile Strength	
	% Change	Requirement	% Change	Requirement
100% Mineral Oil	0.05	0.5% max.	-5.00	+ 12% max.
5% t-Butyl Mercaptan in Mineral Oil	0.04	0.5% max.	-4.91	+ 12% max.
100% Ethylene Glycol	0.01	0.5% max.	-2.54	+ 12% max.
15% Toluene in Methanol	0.01	1.0% max.	-1.94	+ 12% max.

Certifications



Certified as meeting requirements of Plastics Pipe Institute and NSF International, as stated above.

Company Information

For further information regarding the LyondellBasell company, please visit <http://www.lyb.com/>.

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